

Mohamed Mostafa

Software Engineer

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[LinkedIn](#) · [Portfolio](#) · [GitHub](#)

SUMMARY

Backend Engineer with 3+ years of building and operating production payment systems and real-time services in Node.js and TypeScript. Shipped multi-provider payment integrations (PayMob, Amazon Payment Services, Stripe), event-driven background processing, and SSE/WebSocket notifications across two live marketplace platforms. Focused on reliability, idempotency, and performance under real transaction volume.

EXPERIENCE

Software Engineer II | [Lesoll](#) Jun 2023 – Present · Cairo, Egypt · On-site

- Leading backend architecture and development for large-scale hospitality platforms.
- Built and operated the payment system integrating PayMob and Amazon Payment Services, processing **~10K transactions/month** with HMAC-verified webhooks and idempotent handlers that guarantee zero double-charges.
- Developed scalable background processing and real-time communication systems using Redis, BullMQ, and WebSockets.
- Improved API performance and optimized database operations across critical services.
- Delivered core business features, including booking, reporting, CoHost management, and employee tools.
- Managed AWS infrastructure, Docker deployments, and CI/CD pipelines.

PROJECTS

SRVJ — Real-Time Collaborative Whiteboard Platform [Live](#)

- Built real-time collaborative diagram editing using Yjs CRDTs over a custom WebSocket server implementing the Yjs sync and awareness protocols, enabling live cursors, presence, and seamless multiplayer editing.
- Secured WebSocket connections with PASETO v4 authentication and enforced project-level role-based access control, while isolating collaboration sessions into dedicated board rooms.
- Designed a dual-snapshot persistence layer that stores both the authoritative Yjs binary state for lossless recovery and a denormalized JSON representation for efficient querying and API reads.
- Architected the backend using PostgreSQL/Prisma for relational data (users, projects, memberships, sharing) and MongoDB/Mongoose for high-frequency diagram mutations.
- Implemented horizontally scalable notifications using SSE with Redis Pub/Sub fan-out, alongside BullMQ workers for asynchronous email and notification processing.
- Improved platform reliability and security through Zod validation, Helmet, CORS, rate limiting, centralized error handling, structured logging with Pino, and Prometheus metrics. Verified collaboration behavior with integration tests and containerized the platform using Docker Compose behind Nginx.

EDUCATION

Modern Academy – Maadi, Cairo, Jul 2022

Bachelor of Computer Science.

TECHNICAL SKILLS

Backend: Node.js, Express.js, TypeScript, RESTful APIs, WebSockets, SSE.

Databases & Caching: MongoDB, Mongoose, PostgreSQL, Prisma, Redis.

Messaging & Async: BullMQ, message queues, Redis pub/sub.

Payments: PayMob, Amazon Payment Services, Stripe (webhooks, HMAC, idempotency).

Cloud & DevOps: AWS (EC2, EKS, S3, ECR), Docker & Docker Compose, GitHub Actions CI/CD.

Frontend: Vue.js, Vite, TailwindCSS, and UnoCSS.